

IMPERIAL

**Improving Short-Horizon Market-Making with Order
Book and Derivatives Signals: A Machine Learning
Approach**

Applied Project

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Project Specification

The client is an electronic market-making desk in index-related products that continuously quotes two-sided markets in SPY, and its core operational risk at short horizons is adverse selection, the fill that arrives immediately before the price moves against the quote so that the captured spread is eroded or lost. The desk already conditions its quoting on the state of the order book. The open question is whether information from the index options market, where institutional volatility and hedging flow concentrates, would improve those decisions enough to justify integrating a derivatives data feed into the quoting stack.

The research question is fixed as follows: Can derivatives-implied signals improve short-horizon market-making decisions beyond limit order book information alone?

The deliverable is a single data-driven recommendation: yes, no, or under what conditions, supported by a controlled comparison of matched model families tuned under an identical protocol, with the information set as the variable being tested. The horizon is fixed at 10 seconds, a short microstructure horizon that the available data supports. The asset pair is SPY for the order book side and SPX index options for the derivatives side. SPX options are European and cash settled, provide a liquid and widely used index implied-volatility surface, and host the institutional and zero-days-to-expiry flow most likely to contain information.

The committed design is a 2×3 grid in which two information sets, order book only and order book plus derivatives, are crossed with three model families, logistic regression, gradient-boosted trees and a multilayer perceptron. Evaluation is two-tiered, a predictive tier that measures AUC and balanced accuracy on a binary adverse-selection event, and a market-making tier of 10-second markouts and a spread-capture decision rule used to test whether predictive differences translate into economically relevant quote selection. Data are Databento MBP-1 top-of-book records for SPY and OptionMetrics IvyDB end-of-day surface and option-price files for SPX, from January 2024 onwards.

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Project AI Usage and Reflections

Section 1: How I Used AI

I used two generative AI tools during this project. Claude, through Claude Code, was my main assistant for coding, debugging and running the pipeline I designed and specified, while I used ChatGPT separately as an independent reviewer to audit specific parts of the analysis. I fixed the research question, asset universe, ten-second horizon, sample splits, model families, target definition and evaluation metrics myself. Within that design, Claude wrote and debugged code for analyses I had already decided to run, and ChatGPT cross-checked calculations and code logic. Every quantitative result in the report traces to code that I ran and checked against the underlying data.

Section 2: AI-Assisted vs. Non AI-Assisted Work

I designed the experiment without AI, but the implementation would have taken substantially longer without this assistance, given the scale of the raw data and the number of third-party library version issues encountered. Every analytical decision and every line of interpretation is my own, and five cases stand out where I checked a result and acted on what I found.

1. An early pipeline evaluated the final models, fitted on training plus validation data, using scalars fitted on the training window alone. The test results did not match my earlier checks, so I traced the cause, corrected the evaluation code and re-ran the full test-set evaluation, which moved the LOB-only XGBoost pooled AUC from 0.529 to 0.636.
2. The initial analysis showed pooled AUC gains from adding derivatives of +0.008 for logistic regression, +0.021 for XGBoost and +0.038 for the neural network. Before accepting this as the headline finding, I built a placebo in which the derivatives-to-day mapping was permuted, so the added features carried no real information. Permuted noise matched the real model's pooled performance, which meant the result could be reproduced without any genuine derivatives information, and I rejected it.
3. A late cross-check flagged that one supplementary day-level experiment joined option-surface features by their own end-of-day date rather than the previous-trading-day mapping used everywhere else, pairing them with the same day's outcome. I confirmed the join in the code, removed the result from the report and marked it in the notebook as a retired diagnostic, which I preferred to keeping a favourable number I could not defend.
4. Another review flagged that the options quality filter took its cutoff from the whole sample rather than the training window. I found that it changed a single day and, rather than disclose it as a limitation, I moved the cutoff to the training window and re-ran the entire pipeline, because a walk-forward claim should hold exactly.
5. A significant positive intraday estimate appears at 30 minutes in one engineered feature form. I report it in Appendix G with its confidence interval and explain why the clean incremental test at the same horizon is null, rather than omitting a result that cuts against a headline of no robust intraday value.

Section 3: Reflections

I treated AI strictly as an implementation tool inside a research design I had already fixed, and never as a source of research questions, model choices or conclusions, which is what protected academic integrity throughout. The decision to run a placebo test before accepting the derivatives features was mine, as a positive result that cannot survive an information-free control is not a credible finding. I also rewrote the day-level result several times to avoid overstating it. The Diebold-Mariano test on the daily forecast improvement was not significant, but because the two forecasting models are nested that test is conservative in this setting. Therefore, I computed the Clark-West statistic designed for nested comparisons, which was significant at the 5% level on a one-sided test, and I report both.

On the economic side, I withdrew the only positive information-set claim once I established that the decision rule's measured gain tracked how selective each rule was rather than what it knew. Removing a favourable result was right, and so was running a test that strengthened the claim I had been most careful not to overstate, since an evidenced answer is worth more than a convenient one.

Section 4: Additional information you may wish to include

The complete, executable code for every processing and analysis step referenced in this report is available at: <https://github.com/EC152/2026-AP-RMFE>. Excerpts, rather than the full code, are found at the end of the appendices due to the length of the notebook.

Declaration: I acknowledge the use of ChatGPT and Claude. I confirm that no content generated by AI has been presented as my own work.

Date: 12/08/2026

Executive Summary

The desk asked whether adding an SPX derivatives feed would improve 10-second quoting decisions beyond the information already contained in the order book, with the results showing no incremental benefit. Daily derivatives variables add nothing once the SPY order book features are strong, and upgrading those features is the only change that improves prediction on the primary 10-second target. The remaining use for derivatives is forecasting next-day risk, which has statistical evidence but no economic support.

This conclusion comes from a controlled information-set experiment on a 1.1 million interval Databento SPY top-of-book grid merged where available with OptionMetrics SPX end-of-day surface and option-flow features, evaluated out of sample on 193,057 intervals following the April 2025 tariff shock. Adding derivatives features to any of five model classes produces no robust within-day gain, and against the strongest order-book feature set every derivatives variant is significantly negative. Prior-day surface information does improve next-day adverse-selection forecasts, and the Clark-West test for nested models rejects equal predictive accuracy, though the calibrated threshold rules cannot confirm it because their measured value tracks how selective each rule is rather than its predictive information.

1 Introduction

A market maker earns the spread when a passive quote is filled and the price does not move against it before the position can be managed. The key short-horizon risk is adverse selection, the fill that arrives because an informed counterparty expects the price to move through the quote (Glosten & Milgrom 1985). Modern market-making frameworks cast quote placement as a dynamic control problem balancing spread capture, execution and inventory risk (Avellaneda & Stoikov 2008, Guéant et al. 2013). For a desk that already monitors its own order book, the practical question is where the next unit of predictive information comes from. One candidate is the index options market. SPX options concentrate institutional volatility trading, with zero-days-to-expiry contracts averaging 53% of SPX volume in this sample, in line with industry figures (Schwartz 2026), and if options markets carry forward-looking information, their implied features should improve quote-level risk prediction. A derivatives subscription is a recurring cost, and justifying it requires evidence that the signal improves decisions at the horizon where quotes are managed rather than at the horizons the literature has examined. Therefore, the project asks whether derivatives-implied signals can improve short-horizon market-making decisions beyond limit order book information alone.

The project contributes a pipeline joining Databento SPY top-of-book data to OptionMetrics SPX surface and option-flow features on a causal 10-second grid, a test against the pre-committed baseline and a strengthened order-book feature set, and a validation layer of placebo permutation, seed ensembling and within-day decomposition to falsify spurious gains. The evidence rejects a robust 10-second derivatives increment, while predictive value emerges at the day level.

2 Background and Literature

Market-making theory frames what the desk cares about, with Avellaneda & Stoikov (2008) casting quote-setting as a trade-off between spread capture and inventory risk, and Glosten & Milgrom (1985) showing that quoted spreads compensate for trading against better-informed counterparties. Guéant et al. (2013) extend the framework to inventory limits, but none of these models say where the predictive information should come from. The microstructure literature argues it comes from the order book itself. Cont & de Larrard (2013) show that queue imbalance predicts short-horizon price direction and Cont et al. (2014) find that order-flow imbalance relates more robustly to short-interval price changes than raw trade volume, motivating this project’s order-flow features. Sirignano & Cont (2019) document universal price-formation features learnable from order book data at scale, and Kolm et al. (2023) find that models trained on stationary order-flow inputs outperform those trained on raw order-book states.

A separate literature argues that options markets contain information about the underlying, and Easley et al. (1998) model informed traders choosing to trade in options. Empirically, option volume predicts future stock prices (Pan & Poteshman 2006), deviations from put-call parity predict returns (Cremers & Weinbaum 2010), the volatility smirk predicts the cross-section (Xing et al. 2010), and the option-to-stock volume ratio carries information when trade direction is unobserved (Johnson & So 2012), all on daily and weekly horizons. The key counterpoint is Muravyev et al. (2013), who find that option quotes contain no information about future stock prices beyond what current prices already reflect, suggesting the equity market leads at quote-level frequency.

That disagreement is partly methodological, since intraday-lead studies rarely test for the between-day distortions a day-constant predictor can produce in a pooled evaluation. Order-flow variables should dominate at seconds and options variables may matter at slower volatility-regime horizons, so where the boundary lies is the empirical question this project tests at the 10-second horizon, with the day level as the comparison frequency.

3 Data

The order book side uses Databento’s EQUUS.MINI dataset, MBP-1 schema, for SPY, a derived top-of-book feed aggregating proprietary NMS-exchange and ATS feeds into a synthetic NBBO rather than the official SIP NBBO, and it can carry transient crossed or one-sided states. Records are converted to New York time and aggregated to a 10-second grid over the regular session, with each session trimmed to its actual close because the feed carries distorted states after early closes, and with SPY ex-dividend dates excluded. The spread-cap checks in Appendix F confirm that the residual states do not affect the conclusions. The full grid is 1,149,480 intervals over 494 trading days (Table 1). Trades carry the aggressor side where it is reported, so signed order flow requires no classification algorithm, and records with unspecified side contribute zero. Order-count fields are unpopulated, leaving the queue-count imbalance feature identically zero, and descriptive statistics are in Appendix B.

The derivatives side uses end-of-day OptionMetrics IvyDB data for SPX. The surface features are the 30-day at-the-money implied volatility, its first difference, a 25-delta skew and a 30-versus-91-day term spread, joined by near-the-money gamma and vega and by five option-flow features covering put-call implied-volatility spreads, volumes, the zero-days-to-expiry share and open-interest changes. Each day’s surface is deliberately broadcast to every 10-second interval of the following session, testing whether prior-day regime information improves intraday quoting, since intraday implied volatility is unavailable in the academic data. Coverage ends on 2025-08-29, so the September to December 2025 window is used only for order-book-only stability checks. Daily Cboe VIX data serve as a cross-vendor audit, with the OptionMetrics ATM level correlating 0.982 with the VIX at lag zero.

Table 1: Data Sources and Sample Design. Row counts are 10-second bins. The merged sample is smaller because a complete derivatives vector requires both a prior trading day and a first difference, and the split counts exclude the final bin of each trading day, which has no forward markout. The September to December 2025 window supports LOB-only stability checks only.

Sample	Period	LOB-Only Rows	Merged Rows
Full Databento Grid	2024-01-02 to 2025-12-31	1,149,480	
Training	2024-01-02 to 2024-11-29	531,132	526,454
Validation	2024-12-02 to 2025-04-30	232,820	232,820
Primary Test	2025-05-01 to 2025-08-29	193,057	193,057
Robustness (LOB-only)	2025-09-01 to 2025-12-31	191,977	

The sample starts in January 2024 because the zero-days-to-expiry regime that now dominates SPX volume consolidated only recently. The primary experiment uses a single chronological train-validation-test split, with training to November 2024, validation to April 2025, and a primary test window from May to August 2025, the volatile period following the April 2025 tariff shock. Split boundaries fall between trading days and labels do not overlap on the 10-second grid, so no feature or markout window crosses a boundary and purging is unnecessary.

4 Methodology

Features aggregate quotes and trades over the completed interval $[t - 10s, t]$, the decision is taken at the bin close t , and the outcome is the markout over $[t, t + 10s]$, so no feature overlaps the predicted interval (Figure 1). The target is a binary early adverse-selection indicator

$$y_t = \mathbf{1}\left\{q_t(m_{t+10} - m_t) \geq \tau \cdot \frac{1}{2}(a_t - b_t)\right\}, \quad q_t = \text{sign}(\text{signed flow}_t) \quad (1)$$

where m is the mid price and q_t , the direction of net aggressive flow in the completed feature bin, proxies the side on which a passive market maker is exposed.

Since actual maker fills and queue positions are unobserved, y_t is a flow-conditioned proxy for adverse selection rather than a realised maker-fill label. The committed $\tau = 0.5$ flags moves eroding at least 50% of the half-spread, an early-warning definition rather than break-even, with thresholds of 1.0 and 2.0 as robustness checks (Appendix A). When a bin has no net directional signed flow, the exposure side is undefined and $y_t = 0$ by convention. Of the test bins, 85.8% have directional flow, so overall discrimination partly reflects predicting exposure itself, and Section 5.3 shows the conclusion is unchanged on that subset.

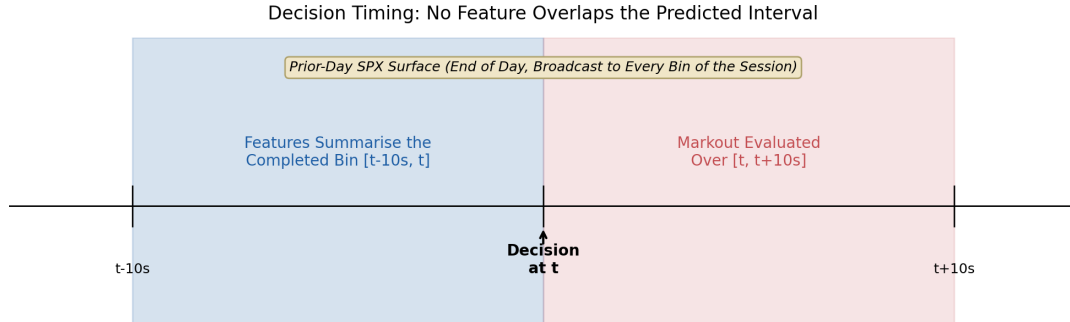


Figure 1: Decision Timing. Features summarise the completed 10-second bin, the decision is taken at the bin close, and the markout is evaluated over the following bin. Derivatives features are the prior day’s end-of-day SPX surface broadcast to every bin of the session.

The experiment compares five feature sets, starting with LOB6, the pre-committed baseline of spread, size imbalance, depth, queue-count imbalance, signed order flow and rolling realised volatility. LOB+ is the strongest constructible order-flow and top-of-book set of nine features, dropping the unpopulated queue feature and adding order-flow imbalance in the sense of Cont et al. (2014), gross traded volume, quote-update intensity and a running measure of session realised volatility. Order-flow imbalance aggregates the signed change in quoted depth at the best quotes, capturing book pressure beyond what trades alone reveal. The derivatives variants add the six surface features, the eleven-feature surface-plus-flow set, or an engineered variance-risk-premium difference, prior-day implied volatility minus running realised volatility.

The model grid crosses two information sets with three families, a class-weighted logistic regression with L2 regularisation, an XGBoost tuned by early stopping, and a multilayer perceptron. Single-seed neural networks gave unstable information-set estimates, so all main MLP results are three-seed ensembles. HistGradientBoosting replicates the comparison in an additional boosting family and a compact temporal convolutional network tests the sequence-model alternative. Logistic regression, XGBoost and the sequence model use class-weighted losses, with the XGBoost weight fixed at its training-window value, while the multilayer perceptron and HistGradientBoosting rely on imbalance-insensitive metrics and thresholding. Appendix C lists final hyperparameters and the pinned environment, and Appendix H maps the analysis notebook.

Hyperparameters are tuned with train-only standardisation, final models are refit on train plus validation, and every matrix those models subsequently touch is transformed with the same scalers. Because derivatives features are constant within a day, pooled AUC also rewards between-day base-rate ranking that day-level predictors can supply, so the primary metric is within-day AUC, the mean of per-day AUCs, checked against a placebo test on permuted derivatives values. Inference on within-day quantities resamples whole test days with replacement, while pooled AUC is descriptive only, since resampling individual intervals would treat dependent observations within a day as independent.

The economic tier uses models fitted on the training window only, so the validation predictions on which isotonic calibration and the decision threshold are selected are out of sample. It evaluates a threshold rule by the improvement in the mean spread-capture proxy among the intervals it retains, relative to quoting every candidate interval. The per-interval proxy is

$$SC_t = \frac{1}{2} (a_t - b_t) - \max\{q_t (m_{t+10} - m_t), 0\} \quad (2)$$

This is the half-spread net of the adverse markout, with favourable moves not credited beyond full retention of the half-spread, before fill probability, queue position, inventory, fee, rebate, hedging and latency effects.

5 Results

5.1 The Locked Grid and the Pooled AUC Distortion

Table 2 reports the pre-committed 2×3 comparison. The non-linear models reach a within-day AUC near 0.61 against 0.505 for logistic regression, so the order book contains predictive signal at 10 seconds. No family shows a robust gain from the derivatives set, since the XGBoost change of -0.0003 is insignificant, the MLP ensemble loses a significant -0.0025 , and HistGradientBoosting is also negative (Appendix D).

Logistic regression shows a large pooled gain of $+0.0369$ that vanishes within day, since a linear model can use day-constant features only as day-level shifts. The placebo test confirms this, as retraining XGBoost with the derivatives day-mapping randomly permuted yields AUCs indistinguishable from the real model, pooled 0.6307 to 0.6327 against 0.6318, so the apparent pooled ranking is reproducible with information-free day-constant features.

Table 2: Predictive Performance of the Locked Grid, Primary Test Window (2025-05-01 to 2025-08-29, $N = 193,057$, adverse base rate 0.388). Within-day AUC is the mean of per-day AUCs. The full eight-model table with confidence intervals is in Appendix D.

Model	Information Set	Pooled AUC	Within-Day AUC
M1 Logistic Regression	LOB Only	0.5155	0.5050
M2 XGBoost	LOB Only	0.6327	0.6103
M3 MLP Ensemble	LOB Only	0.6298	0.6081
M4 Logistic Regression	LOB + Derivatives	0.5524	0.5059
M5 XGBoost	LOB + Derivatives	0.6318	0.6100
M6 MLP Ensemble	LOB + Derivatives	0.6283	0.6055

5.2 The Feature Set Hierarchy: The Central Result

A fair test requires the strongest order book competitor, so Table 3 and Figure 2 estimate one model on five feature sets under an identical protocol. The order book upgrade improves performance, as LOB+ beats the locked baseline by a significant $+0.0033$ within-day AUC and outperforms it on 75% of test days, the largest robust gain at the primary 10-second horizon. No derivatives set beats LOB+, all three are significantly negative against it, and the desk's best 10-second model is therefore order book only.

Table 3: The Feature Set Results Hierarchy. One model, XGBoost under an identical protocol, estimated on five feature sets. All models use the common feature-complete LOB+ sample of 192,642 test rows, so small differences from Table 2 reflect the stricter sample. Gains versus LOB+ carry day-bootstrap confidence intervals.

Feature Set	Pooled AUC	Within-Day AUC	Gain vs LOB+ [95 % CI]
LOB6 (Locked Baseline)	0.6328	0.6103	−0.0033 [−0.0047, −0.0020]
LOB+ (Production Set)	0.6380	0.6136	Benchmark
LOB+ and Surface (6)	0.6367	0.6131	−0.0006 [−0.0010, −0.0001]
LOB+ and Surface + Flow (11)	0.6369	0.6127	−0.0009 [−0.0014, −0.0004]
LOB+ and VRP Difference	0.6375	0.6130	−0.0006 [−0.0009, −0.0003]

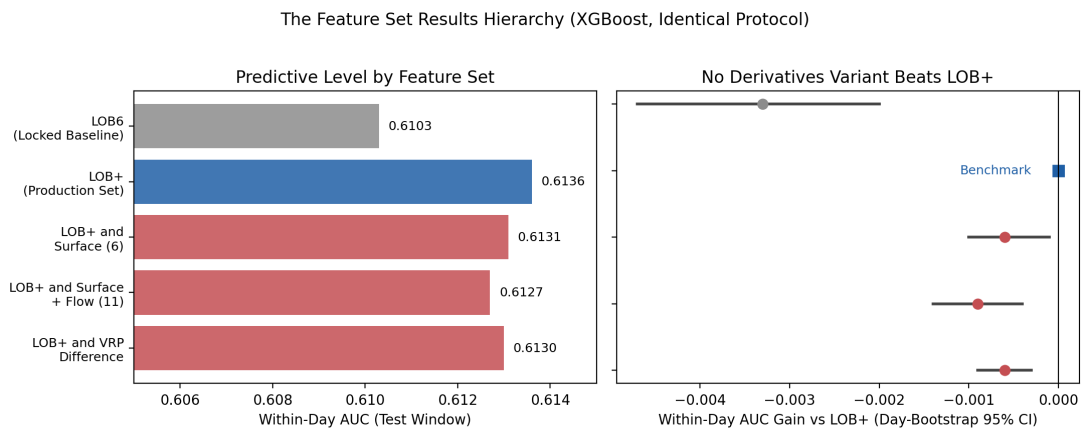


Figure 2: The Information Hierarchy of Table 3. Within-day AUC by feature set on the left and the within-day gain versus the LOB+ benchmark with day-bootstrap confidence intervals on the right.

Feature attribution explains the pattern, since signed order flow dominates by more than an order of magnitude, with mean absolute SHAP 1.09 against 0.049 for the next feature. Gross traded volume is the most important newly added LOB+ feature by gain importance, and the derivatives features collectively receive about 5% of attribution (Figure 3). This is consistent with microstructure evidence that recent order flow and liquidity conditions carry short-horizon price information, and with recent flow summarising much of the relevant history at this horizon.

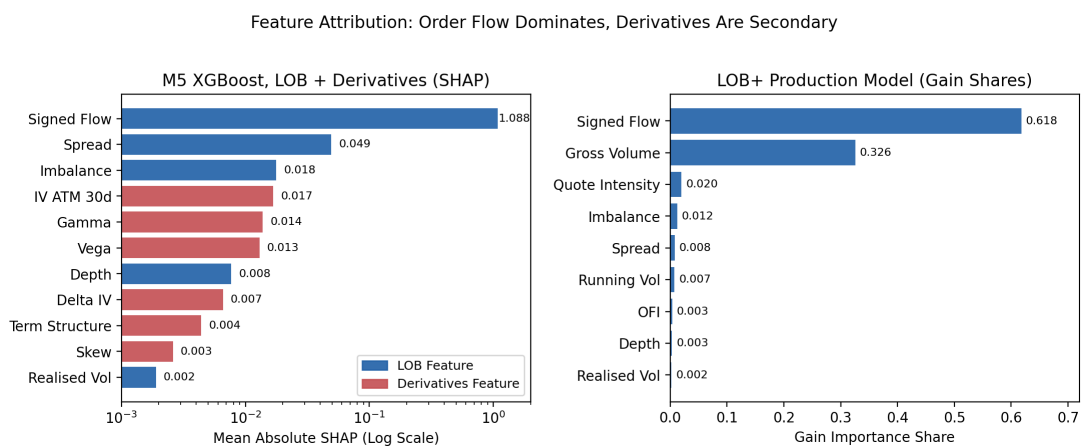


Figure 3: Feature Attribution. Mean absolute SHAP for the derivatives-augmented XGBoost on a log scale with derivatives features in red on the left, and gain-importance shares in the LOB+ production model on the right.

5.3 Robustness and Scope

The primary 10-second conclusion survives the robustness checks in Appendix F, where spread caps of \$0.25 and \$1.00 leave the primary model pair unchanged with 99.9% of rows surviving, so residual one-sided synthetic-NBBO states do not drive the result. Conditioning on bins with directional flow collapses within-day AUC to roughly 0.50 to 0.52 across model classes, so a large share of overall discrimination is exposure prediction, and retraining on active rows leaves an insignificant gain of -0.0011 . Rebuilding the target at the break-even threshold produces no significant gain either, and only the severe threshold shows a small $+0.0021$, though isolated positives among many tests are not conclusive. The information-set sweep to 30 minutes and the attribution to 2 hours find no consistent derivatives benefit, with an isolated positive increment for the surface-plus-flow set at one minute and the remaining comparisons null, negative or unstable (Table 4, Figure 4, Appendix G). The temporal convolutional network ensembles reproduce the null in the sequence-model class, within-day 0.5732 against 0.5598 with derivatives. On the September to December out-of-sample window the locked XGBoost LOB model is stable at 0.7226 within-day AUC, in a regime with a lower adverse-event rate and fewer directional-flow intervals, so exposure prediction is easier.

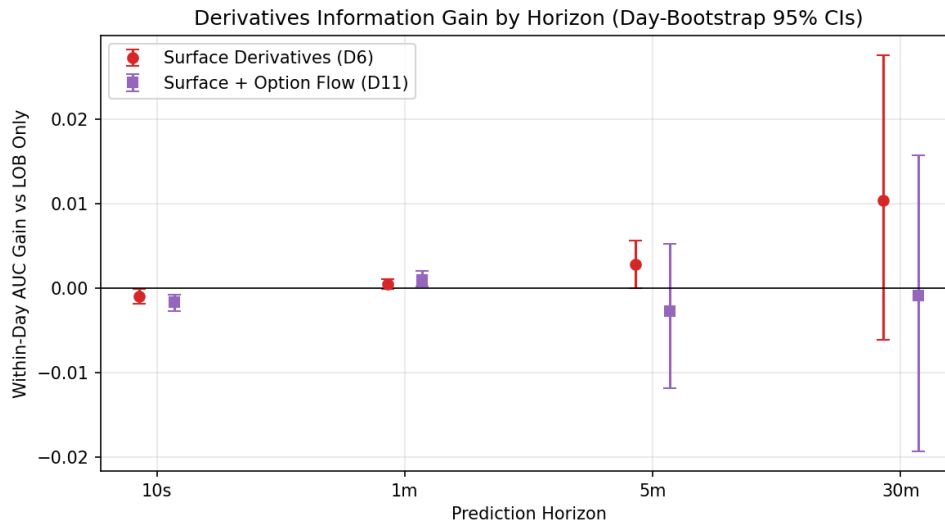


Figure 4: Derivatives Within-Day AUC Increments by Prediction Horizon, with Day-Bootstrap Confidence Intervals.

Table 4: Where derivatives-implied information has value. The intraday rows are within-day AUC increments over the relevant LOB baseline. The day-level row is out-of-sample R^2 against an expanding historical-mean benchmark for next-day adverse-selection rate. The decision-layer row summarises the rule evidence of Table 5, which cannot identify an information-set effect.

Frequency	Derivatives Increment	Evidence
10 s to 5 min	No Consistent Increment	Isolated $+0.0010$ at 1 min, not replicated
30 minutes	No Robust Benefit	Sweep not significant, clean IV increment null, difference-form estimate unstable
2 hours	No Robust Benefit	Adding IV beyond running volatility not significant
1 day	Positive, Nested Test Significant	OOS R^2 0.30 to 0.44, Clark-West one-sided $p = 0.021$
Decision Layer	Not Identified	Rule uplift tracks coverage, not information

5.4 The Decision Layer and the Day Level

The market-making tier flags the next interval as high risk when the isotonic-calibrated probability exceeds a threshold selected on out-of-sample validation predictions from train-only models, subject to a 60% validation flag-rate cap (Appendix E). Every non-degenerate rule improves the mean spread-capture proxy among the intervals it retains (Table 5), but the pairwise derivatives differences do not identify an information-set effect. The statistic tracks selectivity, and across the seven non-degenerate rules the uplift is almost perfectly ordered by flag rate, so a difference between rules at different operating points cannot be attributed to the information set. The exploratory day-level evidence rests on the forecasting channel, where the prior-day derivatives features lift the expanding-window R^2 for next-day adverse-selection forecasts from 0.296 with own order-book day lags to 0.444 with derivatives added, an improvement that is surface-driven. The comparison is nested, so the appropriate test is the adjusted statistic of Clark & West (2007), which rejects equal predictive accuracy on 209 days (one-sided $p = 0.021$) where the unadjusted test does not ($p = 0.115$). Derivatives also forecast next-day volatility better than the desk’s own day lags, 0.605 against 0.571 (Appendix F).

Table 5: Calibrated Decision Rule, ordered by flag rate. Flag Rate is the share of test intervals flagged as high risk, so retained coverage is one minus that share, and Hit Rate is the share of test days with positive conditional uplift. Differences between rules reflect operating point rather than information, so no pairwise information-set comparison is reported. M1 is omitted, its rule flagging essentially all intervals.

Rule	Flag Rate	Δ Proxy, Retained vs All	Day-Bootstrap 95 % CI	Hit Rate
M4 (LR, LOB+D)	48 %	\$0.00300	[\$0.00136, \$0.00464]	51 %
M2 (XGBoost, LOB)	49 %	\$0.00493	[\$0.00412, \$0.00561]	96 %
M7 (HistGB, LOB)	53 %	\$0.00547	[\$0.00456, \$0.00621]	94 %
M8 (HistGB, LOB+D)	58 %	\$0.00600	[\$0.00468, \$0.00705]	89 %
M6 (MLP Ens, LOB+D)	66 %	\$0.00928	[\$0.00816, \$0.01020]	93 %
M3 (MLP Ens, LOB)	66 %	\$0.00899	[\$0.00783, \$0.01000]	95 %
M5 (XGBoost, LOB+D)	68 %	\$0.00866	[\$0.00684, \$0.01012]	84 %

6 Discussion and Recommendation

The negative result is the contribution, since the apparent pooled AUC gains of up to four points seen in early runs were dismantled by a validation layer built to falsify them. The placebo matched pooled performance using information-free features, seed ensembling showed the neural-network gain was unstable across initialisations, and the within-day decomposition separated intraday discrimination from between-day ranking. Adding day identity as a control exposed a scaling error in the final evaluation (Appendix H), and several data-integrity fixes shrank the remaining derivatives value. Fitting the decision layer strictly out of sample then left an economic difference that tracks selectivity, not information. Successive corrections moved the estimate towards the null, showing how easily broadcasting low-frequency features onto a high-frequency grid can produce spurious findings.

The findings help define the boundary between options-information effects and microstructure prediction, with the day-level improvement consistent with Pan & Poteshman (2006) and Cremers & Weinbaum (2010) and the quote-level null consistent with Muravyev et al. (2013). At the 10-second horizon the observed order book and order-flow features dominate the daily SPX derivatives variables, and the best incremental investment is better order-flow modelling. The clearest positive evidence for derivatives information appears at the day level.

Table 6 translates the evidence into decisions. The implementation rule is to use the LOB+ XGBoost model as the quote-level adverse-selection engine and to keep daily SPX derivatives variables out of the quoting stack. Day-ahead conditioning on the surface has informational support but should be validated on the desk’s own fills, since the calibrated rules cannot confirm its economic value.

Table 6: Desk Recommendation Summary. All economic quantities are markout-based spread-capture proxies, not realised PnL.

Desk Decision	Recommendation	Evidence
Add daily SPX derivatives variables to the 10-second quote model	No	No robust within-day AUC gain in any model class or feature form (Tables 2 and 3, Appendix D), and the null holds conditional on directional flow
Invest in LOB and order-flow feature engineering	Yes	LOB+ beats the locked baseline (+0.0033, significant), the largest robust gain at the primary horizon
Use derivatives for day-level threshold conditioning	Conditional Yes	The nested-model test rejects equal predictive accuracy, but the raw comparison is inconclusive and the calibrated rules cannot confirm it because their uplift tracks coverage (Table 5)
Buy an intraday options feed	Future Work Only	The end-of-day surface cannot test a true intraday options signal. OPRA-level flow is the right next experiment

Table 7 records the limitations and how each is controlled. The two that most affect interpretation are the proxy nature of the economic tier and the target convention, under which overall AUC partly reflects exposure prediction. The active-flow analysis shows the conditional problem is hard for every information set, reinforcing the null where it matters most.

Table 7: Limitations and Mitigations

Limitation	Mitigation
OptionMetrics is end-of-day, not intraday	Interpreted throughout as a prior-day regime feature, never as a real-time signal, and the design tests exactly that interpretation
Top-of-book (L1) data only, no deeper-book depth or queue position	LOB+ adds order-flow imbalance, gross volume and quote intensity. Full-depth data is future work
No fill-probability, queue-priority or inventory model	All dollar figures are mean spread-capture-proxy improvements among retained intervals, before fill, queue, inventory, fee, rebate, hedging and latency effects
SPX options map to SPY ETF risk state, not the same instrument	Economically natural index and ETF pair. Features proxy the index-level volatility and skew environment
Target labels non-directional-flow bins as non-adverse	Disclosed, and the null holds under active-flow-only retraining on the economically exposed subset
Derivatives sample ends 2025-08-29	September to December 2025 used for LOB-only model stability, never for information-set comparisons
Hyperparameters tuned on pooled validation AUC	The same ex-ante protocol is applied to both information sets, though pooled tuning could in principle favour day-constant columns. Final inference uses within-day AUC, and metric-aligned retuning is an untested extension

7 Conclusion

Can derivatives-implied signals improve short-horizon market-making decisions beyond limit order book information alone? At the primary 10-second target the answer is robustly no. Across five model classes, multiple derivatives feature constructions and every validation check applied, no stable within-day gain exists, and no derivatives-augmented alternative outperforms the strongest order book feature set, which beats every one of them outright. At the day level the evidence is positive. Prior-day SPX surface information improves next-day adverse-selection forecasts and the nested-model test rejects equal predictive accuracy, though the calibrated threshold rules cannot confirm it. This answer survived a deliberately adversarial validation layer of placebo permutation, multi-seed ensembling and within-day decomposition, so the null at the quote horizon does not depend on a single model family, seed or feature specification.

The lesson for the desk is to prioritise top-of-book and order-flow feature engineering for immediate quote decisions and to treat derivatives-implied variables as a day-level risk input pending economic validation. The methodological lesson generalises, since pooled metrics can overstate day-constant features, and placebo tests, seed ensembles and within-day decomposition should be standard practice wherever a slowly updating signal is merged onto a much higher-frequency panel. Future work should test intraday options flow at the OPRA level, which would separate a real-time options signal from the end-of-day broadcast effect, alongside full-depth order book data and a fill-aware evaluation whose spread capture is actually achievable.

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Appendix A Label-Threshold and Tau Sensitivity

The target multiplier τ and the flagging threshold of the economic rule were both varied. Appendix F Panel B reports the target rebuilt at the break-even (1.0) and severe (2.0) half-spread multipliers. The information-set null holds at break-even and the severe threshold shows only a small positive gain. For the non-degenerate rules the flagging-threshold sweep keeps the gain positive over a broad range around the selected threshold, falling to zero as the flag rate declines, so it does not depend on the exact threshold.

Figure 5 shows the sweep for the spread-capture gain and the flag rate of all eight calibrated rules.

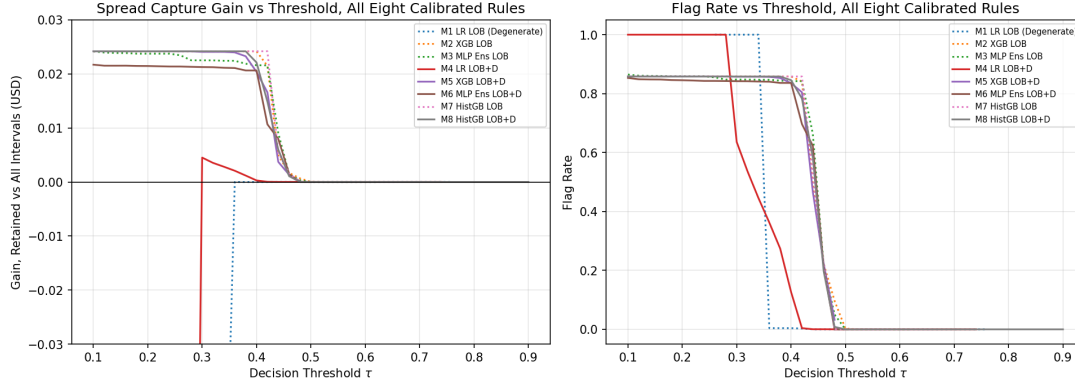


Figure 5: Threshold Sensitivity of the Calibrated Rules. Robust spread-capture gain per model as the flagging threshold varies on the outlier-robust primary test subset, a sensitivity check rather than the threshold-selection mechanism, which uses validation only. The τ on the horizontal axis is this flagging threshold, not the target multiplier.

Appendix B Descriptive Statistics

Table 8 reports feature-level descriptive statistics on the primary test window. A few one-sided synthetic-NBBO states remain in the predictive sample, 7 rows with spread above \$2, all mid-morning, and the economic tier excludes the first session minute and spreads above \$1 throughout. Approximately 29% of trade records carry an unspecified aggressor side and contribute zero to signed flow. Of the test bins, 85.8% carry directional signed flow, 12.4% carry traded volume with no net direction, and 1.8% carry no traded volume.

Table 8: Feature Descriptive Statistics, Primary Test Set ($N = 193,057$). `queue_imb` is identically zero (unpopulated in EQU.S.MINI). `spread` in USD, `vega` in USD per unit change in implied volatility, so a one percentage point move is one hundredth of the tabulated value, and `realised_vol` is the rolling 30-tick standard deviation of mid returns at 10^{-5} scale.

Feature	Mean	Std	Min	25%	Median	75%	Max
<code>spread</code>	0.0235	0.0370	0.0000	0.0100	0.0200	0.0300	7.3900
<code>imbalance</code>	-0.008	0.4626	-0.999	-0.367	0.0000	0.3426	0.9990
<code>depth</code>	930.9	610.6	5.0	528.0	790.0	1216.0	22536.0
<code>queue_imb</code>	0.000	0.000	0.000	0.000	0.000	0.000	0.000
<code>signed_flow</code>	-11.4	1278.5	-167943	-196	0.0	200.0	35191.0
<code>realised_vol</code> ($\times 10^{-5}$)	0.5	2.3	0.0	0.2	0.4	0.7	315.1
<code>iv_atm_30</code>	0.1418	0.0229	0.1060	0.1264	0.1386	0.1487	0.2083
<code>delta_iv</code>	-0.001	0.0109	-0.036	-0.006	-0.001	0.0034	0.0267
<code>skew</code>	0.0521	0.0114	0.0355	0.0431	0.0486	0.0598	0.0821
<code>term_structure</code>	-0.013	0.0052	-0.022	-0.017	-0.013	-0.010	0.0019
<code>gamma</code>	0.0012	0.0001	0.0010	0.0012	0.0013	0.0013	0.0015
<code>vega</code>	1338.5	81.7	1150.1	1281.1	1362.0	1401.8	1486.4

Figure 6 shows the feature correlation structure, where the LOB features are mostly weakly correlated, apart from a moderate association between spread and realised volatility, while the derivatives block contains a strongly correlated core of level, skew, term structure and gamma, with the daily change in implied volatility and vega standing further apart.

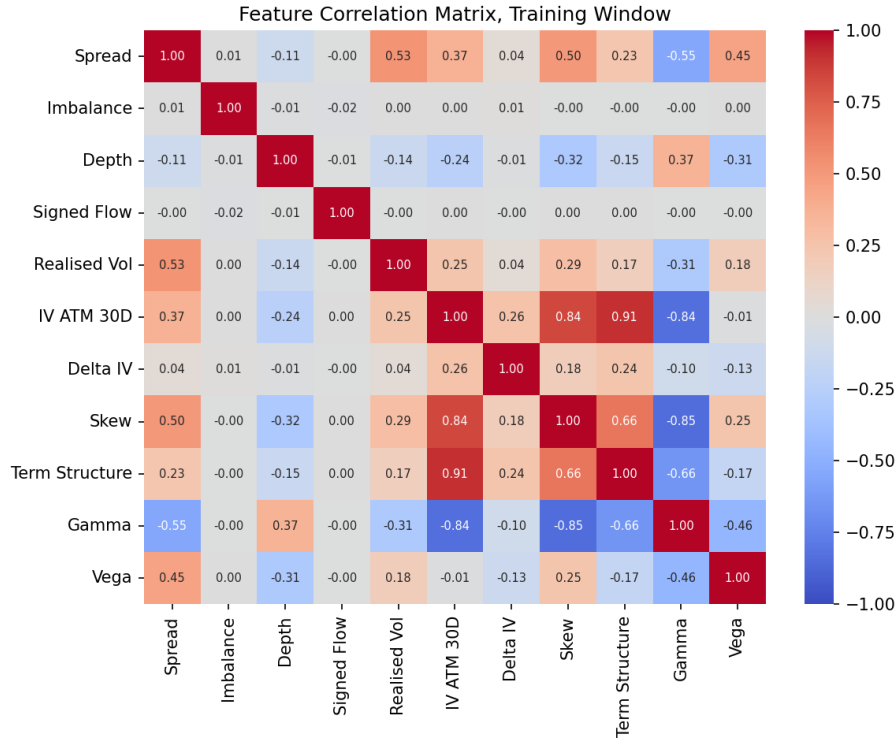


Figure 6: Feature Correlation on the Training Window.

Appendix C Hyperparameters and Reproducibility

Table 9 lists the validation-selected configurations of the eight tabular models M1 to M8.

Table 9: Final Model Hyperparameters. Configurations are tuned on validation and re-trained on the combined train plus validation window, all under the pinned environment (xgboost 2.1.4) with consistent train+val scaling for final evaluation.

Config	Hyperparameter	Value
M1 LR	Regularisation C	10.0
M4 LR	Regularisation C	0.01
M2 XGB	max_depth / learning_rate	3 / 0.1
	subsample / n_estimators	0.8 / 74 (early stopping)
M5 XGB	max_depth / learning_rate	3 / 0.1
	subsample / n_estimators	0.8 / 90 (early stopping)
M3 MLP	L2 α	10^{-4} , ensemble of seeds 42, 7, 123
M6 MLP	L2 α	10^{-4} , ensemble of seeds 42, 7, 123
M3/M6	Architecture	(128, 64), ReLU, early stopping, validation_fraction=0.05, n_iter_no_change=7
M7 HistGB	learning_rate / max_leaf_nodes	0.1 / 31
M8 HistGB	learning_rate / max_leaf_nodes	0.05 / 31
M7/M8	max_iter	300 (early stopping)

The core modelling pipeline uses fixed random seeds (42, with 7 and 123 for ensembles) under a pinned environment: Python 3.10.8, numpy 2.2.6, pandas 2.2.2, scikit-learn 1.5.0, xgboost 2.1.4, statsmodels 0.14.2, databento 0.79.0, torch 2.12.0 (CPU). Inputs are the raw Databento daily files, the OptionMetrics parquet extracts and the Cboe VIX file. Outputs are per-day feature files, the merged grid, saved test predictions and the quantitative inputs to the tables and figures in this report. Saved predictions are reproducible under the pinned environment and fixed random seeds and every preprocessing step is walk-forward. The OptionMetrics surface is quality filtered by dropping the top 5% of the vendor dispersion field, and that percentile is estimated on the training window alone and then applied unchanged to validation and test. The notebook reports the sensitivity of that choice. The training-window cutoff is 0.0524 against 0.0494 on the full sample, so the final filter is the looser of the two, and exactly one of the 1,668 surface points the features read lies between them, a 25-delta put on 2025-04-08. That single point adds one day to the derivatives panel, 416 against 415, and leaves every shared day's features identical and the test window untouched.

Appendix D Full Model Grid and Conditioning

Table 10 reports the full eight-model grid with confidence intervals and Table 11 the information-set gains with inference. Intraday discrimination gains are insignificant or negative in every family. Pooled differences are shown for completeness but carry no inference here, since the pooled metric is the one this project shows to be distorted by day-constant predictors. Figure 7 decomposes pooled against within-day AUC, Figure 8 shows the pooled ROC curves, and Figure 9 tracks weekly within-day AUC through the test window, which varies substantially across weeks with no persistent derivatives divergence.

Table 10: Predictive Performance for All Eight Models, Primary Test Set ($N = 193,057$). Pooled AUC is a descriptive point estimate, within-day AUC carries day-bootstrap confidence intervals, and balanced accuracy uses a threshold selected on out-of-sample validation predictions from train-only models. M3/M6 are three-seed MLP ensembles.

Model	Information Set	Pooled AUC	Within-Day AUC	95 % CI	Bal. Acc.
M1 (LR)	LOB Only	0.5155	0.5050	[0.5016, 0.5085]	0.4993
M2 (XGB)	LOB Only	0.6327	0.6103	[0.5853, 0.6390]	0.6157
M3 (MLP ens)	LOB Only	0.6298	0.6081	[0.5832, 0.6368]	0.6144
M4 (LR)	LOB + Deriv	0.5524	0.5059	[0.5024, 0.5094]	0.5131
M5 (XGB)	LOB + Deriv	0.6318	0.6100	[0.5851, 0.6390]	0.6157
M6 (MLP ens)	LOB + Deriv	0.6283	0.6055	[0.5804, 0.6342]	0.6140
M7 (HistGB)	LOB Only	0.6334	0.6109	[0.5856, 0.6397]	0.6157
M8 (HistGB)	LOB + Deriv	0.6293	0.6093	[0.5839, 0.6384]	0.6157

Table 11: Information-Set Gains from Adding Derivatives, Primary Test Set. Pooled Δ AUC is descriptive. Within-day Δ AUC is the mean of per-day differences with day-bootstrap confidence intervals and carries the inference, and Days > 0 is the share of test days on which the gain is positive.

Comparison	Pooled Δ AUC	Within-Day Δ AUC [95 % CI]	Days > 0
M1 $\rightarrow$ M4 (LR)	+0.0369	+0.0008 [−0.0012, +0.0027]	55 %
M2 $\rightarrow$ M5 (XGB)	−0.0009	−0.0003 [−0.0011, +0.0004]	45 %
M3 $\rightarrow$ M6 (MLP)	−0.0015	−0.0025 [−0.0044, −0.0007]	35 %
M7 $\rightarrow$ M8 (HistGB)	−0.0041	−0.0015 [−0.0029, +0.0000]	37 %

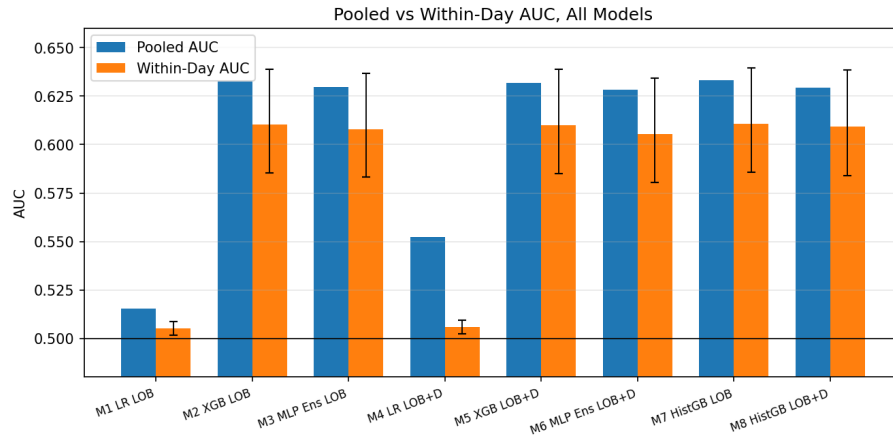


Figure 7: Pooled Versus Within-Day AUC by Model. The gap illustrates the sensitivity of pooled AUC to between-day ranking, which a day-level predictor can capture even without within-day discrimination.

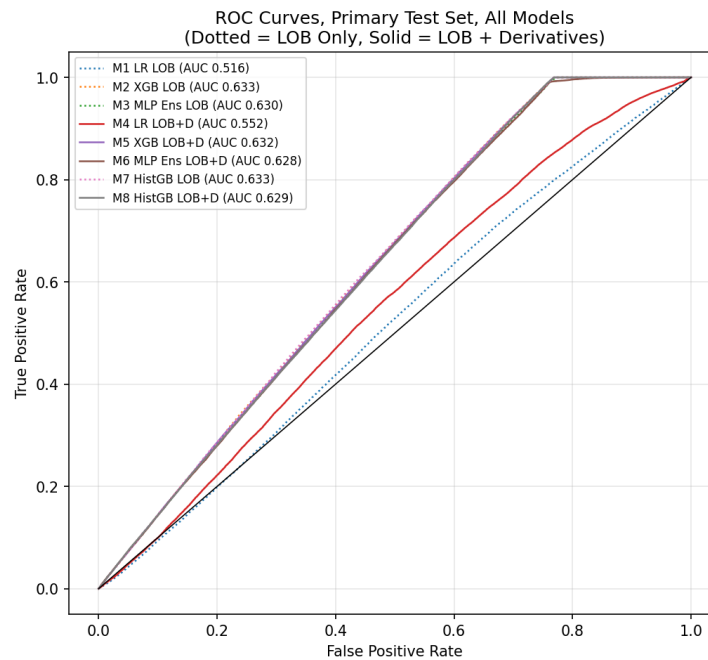


Figure 8: Pooled ROC Curves for All Eight Models, Primary Test Window.

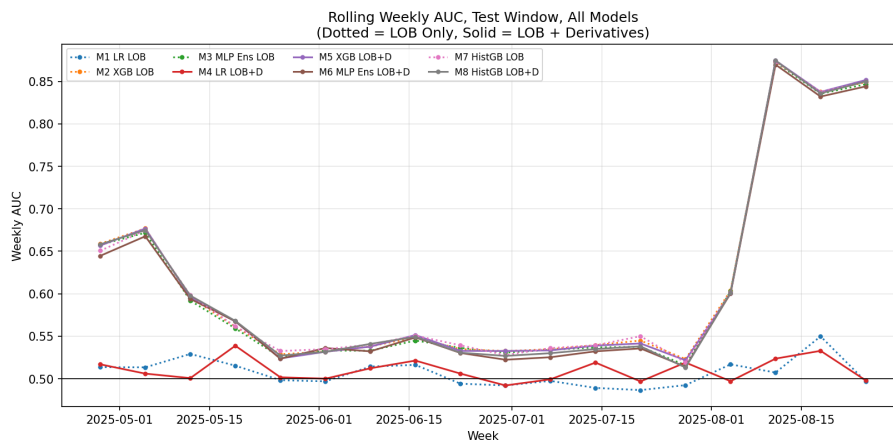


Figure 9: Rolling Weekly Within-Day AUC on the Test Window.

Figure 10 extends feature attribution to every configuration with the method appropriate to each family, absolute standardised coefficients for the linear models, mean absolute SHAP for the boosted trees and permutation AUC drop for the MLP ensembles. The linear LOB+Derivatives panel loads visibly on derivatives columns because a linear model can use day-constant features only as day-level shifts, which is exactly the between-day placement visible in the pooled metrics. Figure 11 shows the SHAP distributions for the boosted configurations and Table 12 the corresponding mean absolute SHAP values for M5.

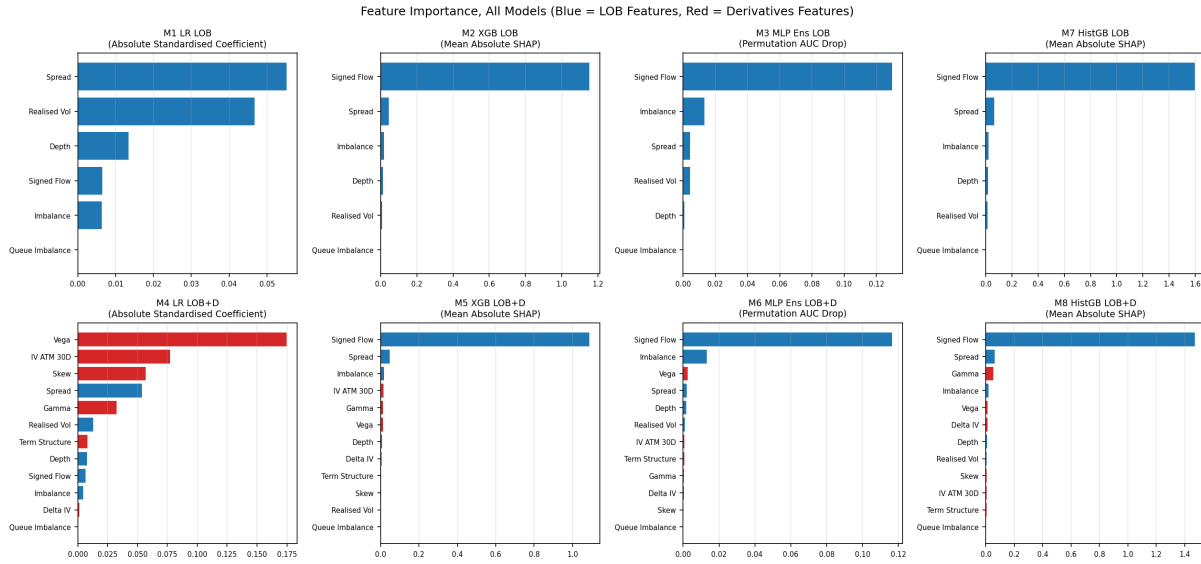


Figure 10: Feature Importance for All Eight Models by the Method Appropriate to Each Family.

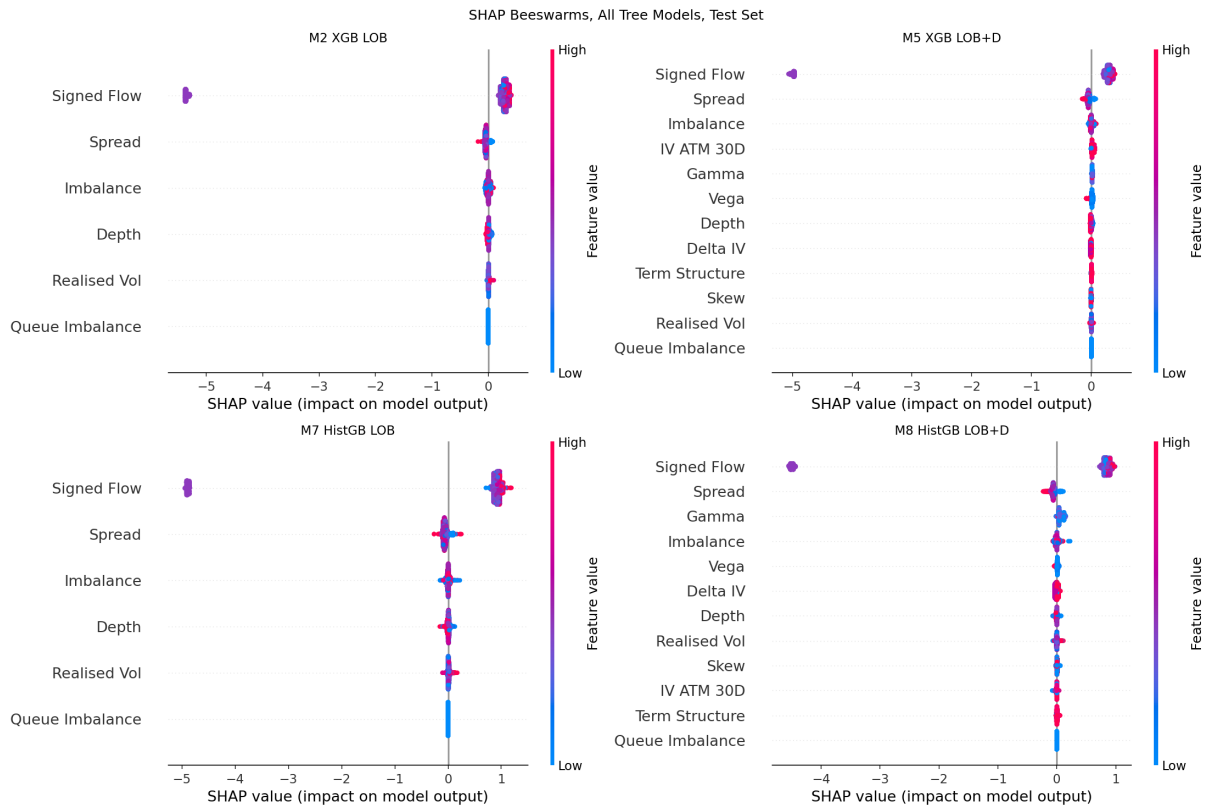


Figure 11: SHAP Beeswarm Distributions for the Four Boosted Configurations.

Table 12: Mean Absolute SHAP for the Derivatives-Augmented XGBoost (M5) (5,000 Observations).

Feature	Mean SHAP	Group
signed_flow	1.0884	LOB
spread	0.0492	LOB
imbalance	0.0177	LOB
iv_atm_30	0.0168	Derivatives
gamma	0.0138	Derivatives
vega	0.0130	Derivatives
depth	0.0077	LOB
delta_iv	0.0066	Derivatives
term_structure	0.0044	Derivatives
skew	0.0026	Derivatives
realised_vol	0.0019	LOB
queue_imb	0.0000	LOB

Table 13 shows no systematic regime or time-of-day segment of derivatives value, LR’s open and close entries reflect day-level placement rather than intraday signal, and the null holds inside both latent volatility regimes. Table 14 reports the out-of-sample window, where the non-linear models generalise strongly, with a lower adverse base rate of 0.317 against 0.388 and active-flow share of 0.69 against 0.86 despite higher realised volatility, 0.68 against 0.52 at the 10^{-5} scale. Fewer directional-flow bins can lift AUC there by raising the share of mechanically safe observations, so the window reflects model stability, not evidence for derivatives augmentation. The latent state of Figure 12 agrees with the IV-median split on 72% of test days.

Table 13: Conditioning Analyses, Primary Test Set. Panel A splits the within-day derivatives gain by day-level IV regime (median split, 41 high-IV and 42 low-IV days), Panel B reports pooled gains by session bucket, and Panel C conditions on the ex-post latent volatility state of Figure 12.

<i>Panel A: Within-Day ΔAUC by IV Regime</i>				
	M1 $\rightarrow$ M4	M2 $\rightarrow$ M5	M3 $\rightarrow$ M6	M7 $\rightarrow$ M8
High-IV Days	+0.0015	+0.0003	−0.0014	−0.0002
Low-IV Days	+0.0001	−0.0009	−0.0036	−0.0028
<i>Panel B: Pooled ΔAUC by Session Bucket</i>				
	M1 $\rightarrow$ M4	M2 $\rightarrow$ M5	M3 $\rightarrow$ M6	M7 $\rightarrow$ M8
Open 09:30 to 11:00	+0.0413	−0.0001	−0.0022	−0.0032
Midday 11:00 to 14:00	+0.0277	−0.0012	−0.0022	−0.0039
Close 14:00 to 16:00	+0.0405	−0.0004	+0.0008	−0.0032
<i>Panel C: Within-Day ΔAUC by Latent Volatility Regime (Markov Switching)</i>				
	M1 $\rightarrow$ M4	M2 $\rightarrow$ M5	M3 $\rightarrow$ M6	M7 $\rightarrow$ M8
High-Volatility Days	+0.0015	+0.0004	−0.0017	−0.0006
Low-Volatility Days	+0.0005	−0.0006	−0.0029	−0.0019

Table 14: Out-of-Sample Stability. LOB-only models applied to September to December 2025 ($N = 191,977$, unseen, scalars not refit). LOB+ is evaluated on its feature-complete subset of 191,562 rows.

Model	Pooled AUC	Within-Day AUC
M1 (LR)	0.5099	0.5139
M2 (XGB)	0.7318	0.7226
M3 (MLP ens)	0.7298	0.7209
M7 (HistGB)	0.7324	0.7235
LOB+ (XGB)	0.7372	0.7251

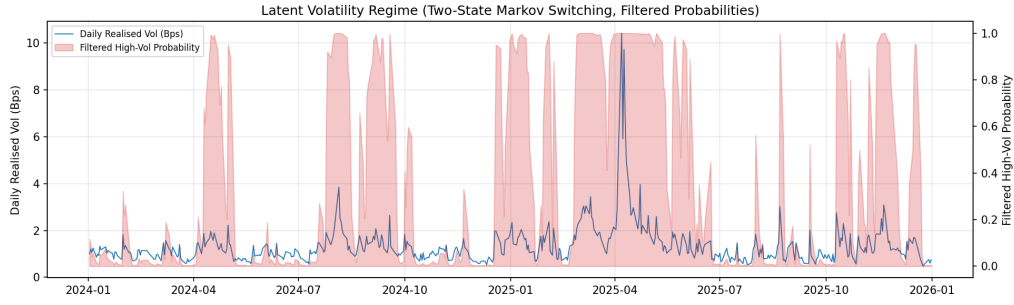


Figure 12: Latent Volatility Regime. Ex-post filtered probabilities from a two-state Markov-switching model estimated on the full sample of daily log realised volatility.

Appendix E Calibration and Economic Detail

Table 15 reports calibration quality for the economic-tier models, with reliability curves in Figure 13 and raw probability distributions in Figure 14. The isotonic map calibrates the boosted models well, with the M2 ECE falling from 0.096 to 0.0004, the HistGB models are already well calibrated raw, and the logistic configurations remain poorly calibrated after mapping. The calibrated probabilities concentrate near the 0.39 base rate, which is what the signal amounts to at 10 seconds and why decision thresholds must be set economically rather than at 0.5.

Table 15: Probability Calibration, Primary Test Set. Isotonic regression is fitted on out-of-sample validation predictions from train-only models. Brier is the mean squared probability error and ECE the expected calibration error over 10 bins.

Model	Brier Raw	Brier Iso	ECE Raw	ECE Iso
M1 (LR)	0.2505	0.2388	0.1134	0.0368
M2 (XGB)	0.2230	0.2123	0.0964	0.0004
M3 (MLP ens)	0.2144	0.2135	0.0133	0.0033
M4 (LR)	0.2402	0.2386	0.0631	0.0516
M5 (XGB)	0.2232	0.2126	0.0959	0.0047
M6 (MLP ens)	0.2182	0.2153	0.0448	0.0074
M7 (HistGB)	0.2124	0.2123	0.0035	0.0027
M8 (HistGB)	0.2130	0.2125	0.0098	0.0055

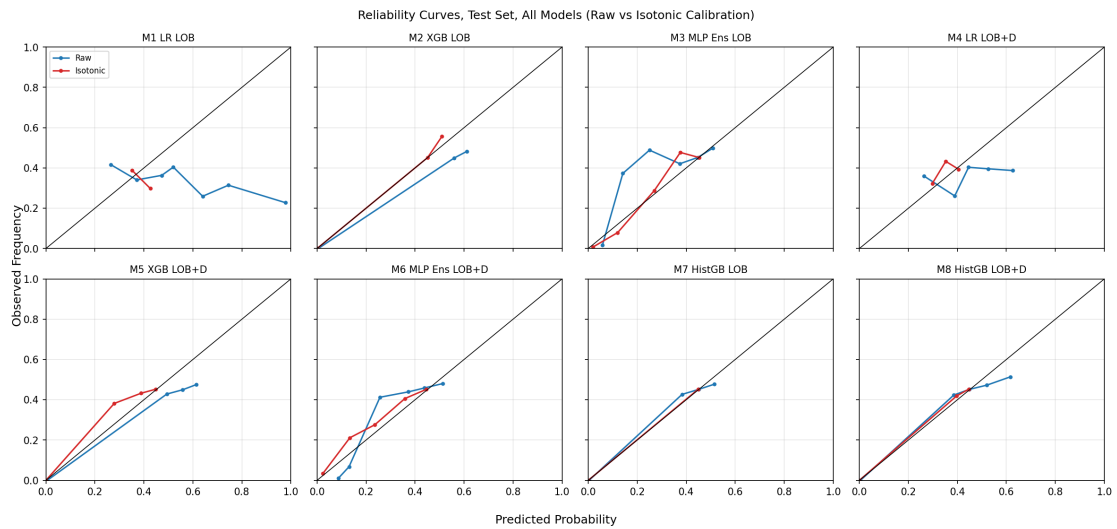


Figure 13: Reliability Curves Before and After Isotonic Calibration.

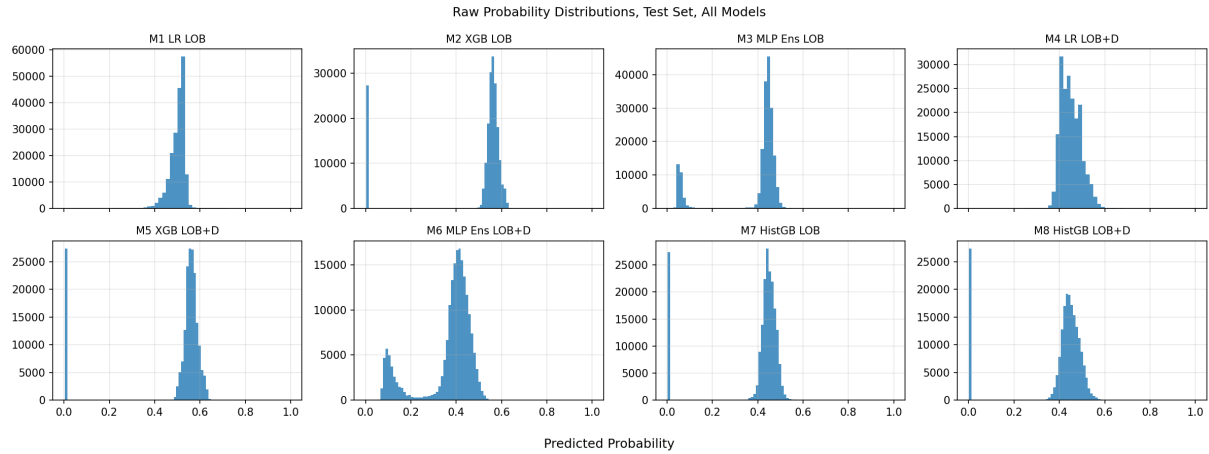


Figure 14: Distributions of Raw Predicted Probabilities by Model.

Table 16 and Figure 15 report the decile lift. The non-linear models separate cleanly, with an essentially zero observed target-event rate in the bottom decile and a top-decile adverse rate of 0.47 to 0.48 against the 0.388 base rate. The zero rates in decile 1 partly reflect intervals without directional flow, where the event is zero by construction and the models learn to score them lowest.

Table 16: Realised Adverse-Selection Rate by Decile of Predicted Probability, Primary Test Set.

Decile	M1	M2	M3	M4	M5	M6	M7	M8
1	0.350	0.000	0.000	0.251	0.000	0.005	0.000	0.000
2	0.361	0.241	0.253	0.326	0.243	0.255	0.239	0.245
3	0.364	0.426	0.429	0.383	0.428	0.425	0.427	0.436
4	0.395	0.439	0.434	0.389	0.440	0.433	0.440	0.442
5	0.402	0.447	0.443	0.397	0.444	0.446	0.438	0.446
6	0.414	0.449	0.449	0.428	0.452	0.454	0.452	0.451
7	0.416	0.454	0.462	0.441	0.455	0.456	0.464	0.454
8	0.404	0.474	0.462	0.439	0.465	0.462	0.465	0.458
9	0.398	0.470	0.468	0.437	0.474	0.466	0.474	0.469
10	0.374	0.477	0.478	0.386	0.476	0.476	0.479	0.477

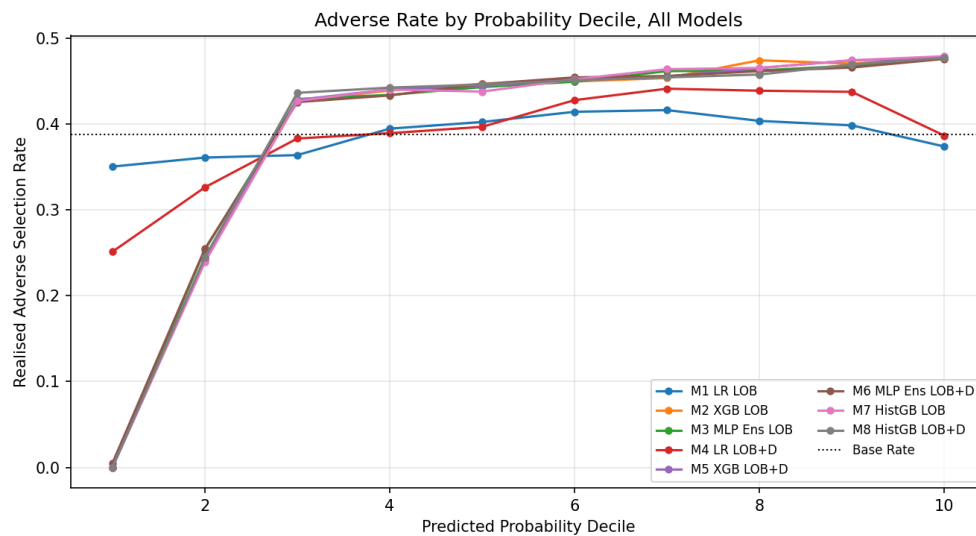


Figure 15: Adverse Rate by Predicted-Probability Decile.

Figure 16 shows the cumulative daily spread-capture improvement of the calibrated rules, the day-weighted view that complements the pooled estimates of Table 5, where differences between rules reflect their different coverage as much as their information.

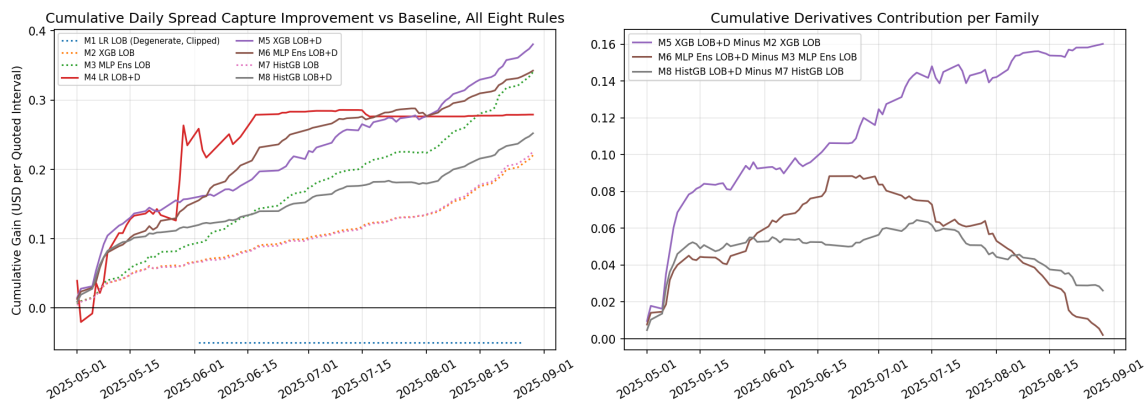


Figure 16: Cumulative Daily Spread-Capture Improvement of the Calibrated Rules on the Test Window. The right panel compares rules operating at different coverage levels and is descriptive only.

Figure 17 maps calibrated probability bins to realised adverse cost relative to the half-spread. The cost shows a broadly increasing relationship with predicted risk for every model with usable calibration, so the model output translates into a multiplicative spread-widening schedule inspired by Avellaneda & Stoikov (2008).

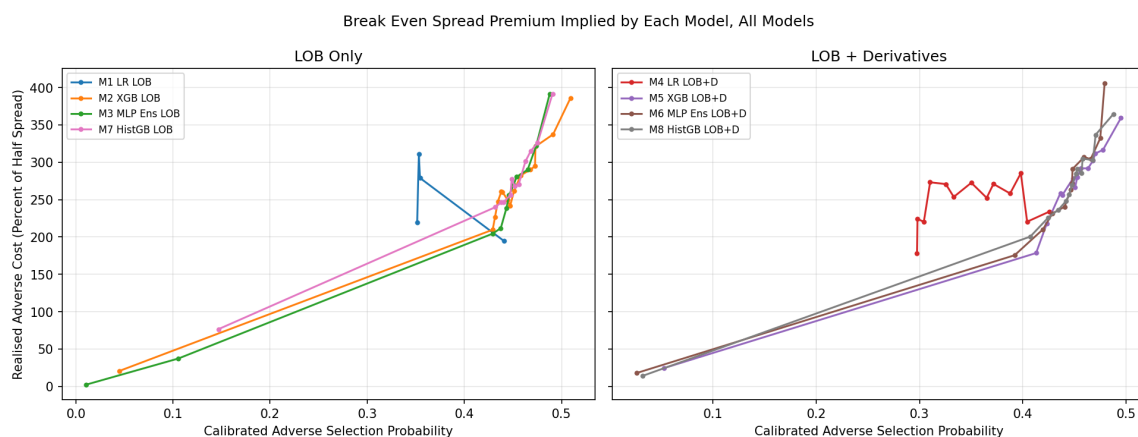


Figure 17: Realised Adverse Cost Relative to the Half-Spread by Calibrated Probability Bin.

Table 17 completes the economic channel. Mean absolute markout rises monotonically with the IV level and quoted spreads widen in the top quintiles, consistent with adverse-selection theory, while the adverse-event rate peaks in the middle quintiles, consistent with the event being defined relative to the half-spread.

Table 17: Spread and Adverse Selection by Implied-Volatility Quintile, Outlier-Robust Test Subset.

IV Quintile	Mean <code>iv_atm_30</code>	Mean Spread (\$)	Adverse Rate	Mean Markout (bps)
1	0.1154	0.0223	0.2555	0.3751
2	0.1297	0.0176	0.4066	0.6447
3	0.1388	0.0179	0.4374	0.6715
4	0.1474	0.0235	0.4370	0.8813
5	0.1795	0.0350	0.4062	1.0008

Appendix F Robustness Detail

Table 18 collects the sample-definition, label-threshold and day-level inference checks. In Panel A, excluding non-positive and wide spreads changes nothing, with 99.9% of rows surviving, while conditioning on bins with directional flow, 86% of rows and the economically exposed case, collapses within-day discrimination to roughly 0.50 to 0.52 across model classes. Therefore, a large share of the overall discrimination is exposure prediction, identifying bins without net directional signed flow that are safe under the target convention, which the calibrated rules exploit by tightening in safe windows. Retraining on active rows does not recover the difference, and the derivatives gain stays negative at -0.0011 but is not significant. In Panel B the null holds at the break-even multiplier, and the severe multiplier shows a small positive gain. In Panel C the day-level forecasting improvement is surface-driven under leave-group-out attribution, but neither the DM test nor the 10-day moving-block bootstrap over the 209 evaluation days is significant.

Table 18: Robustness Checks for Sample Definition, the Activity Convention and the Label Threshold. Panel A recomputes the primary model pair on filtered test subsets. Panel B rebuilds the target at higher half-spread multipliers. Panel C reports inference on the day-level forecasting gain. Panel D reports the placebo and seed checks of the validation layer.

<i>Panel A: Sample Definition (Within-Day AUC, M2 / M5 / Derivatives Gain)</i>			
Sample	M2	M5	Gain [CI]
Full Test Sample	0.6103	0.6100	-0.0003 [$-0.0011, +0.0004$]
Spread > 0 and $\leq \$0.25$	0.6103	0.6100	-0.0003 [$-0.0011, +0.0004$]
Spread > 0 and $\leq \$1.00$	0.6103	0.6100	-0.0003 [$-0.0011, +0.0004$]
Active Flow Only	0.5214	0.5216	$+0.0003$ [$-0.0009, +0.0015$]
Active Flow, Retrained	0.5213	0.5201	-0.0011 [$-0.0024, +0.0001$]
Active-flow within-day AUC by model: M1 0.5038, M2 0.5214, M3 0.5186, M4 0.5051, M5 0.5216, M6 0.5159, M7 0.5219, M8 0.5205			
<i>Panel B: Label Threshold (XGB Pair Retrained per Threshold)</i>			
Threshold	LOB Within-Day	LOB+D Within-Day	Gain [CI]
0.5 (Early Indicator, Primary)	0.6103	0.6100	-0.0003 [$-0.0011, +0.0004$]
1.0 (Break-Even)	0.6118	0.6125	$+0.0007$ [$-0.0002, +0.0017$]
2.0 (Severe)	0.6259	0.6280	$+0.0021$ [$+0.0010, +0.0034$]
<i>Panel C: Inference on the Day-Level Adverse-Rate Forecasting Gain</i>			
OOS R^2 : LOB lags 0.296, plus derivatives 0.444 (surface-driven: dropping the surface removes the gain, dropping option flow improves the forecast)			
DM on daily squared errors (NW lag 5): $+1.57$ ($p = 0.115$)			
Clark-West MSPE-adjusted statistic (NW lag 5): $+2.04$ (one-sided $p = 0.021$)			
Block-bootstrap CI on the mean squared-error improvement: $[-0.00007, +0.00268]$			
Ridge control, expanding OOS R^2 : LOB lags 0.330, LOB lags plus derivatives 0.457			
Next-day volatility OOS R^2 : LOB lags 0.571, derivatives 0.605, combined 0.587			
<i>Panel D: Validation Layer</i>			
Placebo permutations, pooled and within-day AUC: 0.6307 and 0.6095, 0.6327 and 0.6102, 0.6318 and 0.6099, against the real M5 at 0.6318 and 0.6100			
Per-seed MLP within-day derivatives gain: -0.0031 , -0.0061 and -0.0033 , ensemble -0.0025			

Figure 18 shows the improvement accruing in a handful of stress transitions, while the unadjusted test remains inconclusive even though the adjusted test rejects equal predictive accuracy. A ridge-regularised control reproduces the improvement, with an expanding out-of-sample R^2 of 0.330 for the order book day lags and 0.457 with derivatives added, so the gain is not due to unregularised least squares. For next-day volatility the derivatives features also help, with an out-of-sample R^2 of 0.605 for the derivatives block alone and 0.587 for the combination against 0.571 for the order book day lags, so the day-level information is not confined to adverse-selection conditions.

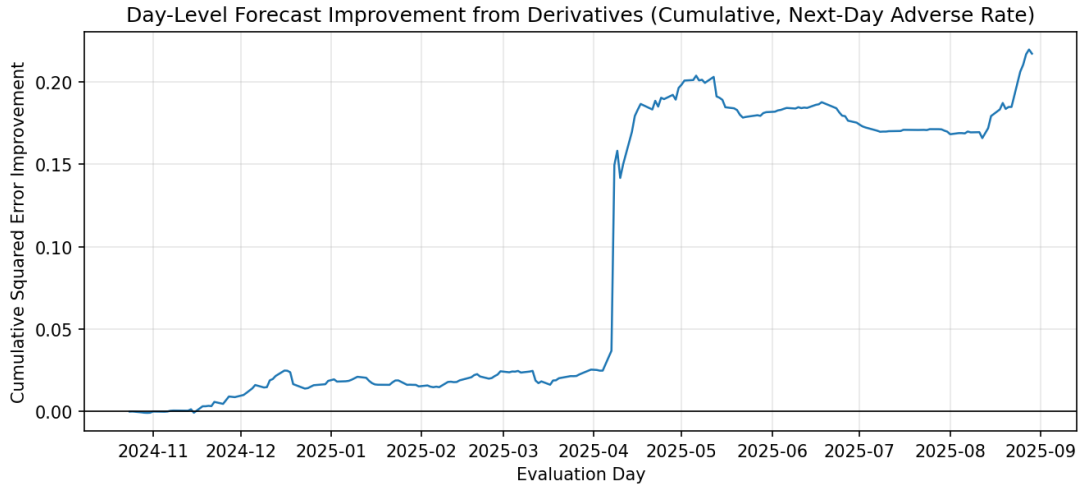


Figure 18: Cumulative Squared-Error Improvement of the Day-Level Forecast from Adding Derivatives.

The day-level comparison is nested, since the augmented model adds the derivatives features to the same order-book day lags. In that setting a test on the raw squared-error difference is undersized, because the larger model carries estimation noise even when the added coefficients are zero in population, so the adjusted statistic of Clark & West (2007) is the appropriate one. It reaches +2.04 with a one-sided p of 0.021 and rejects equal predictive accuracy, while the unadjusted test does not. The difference is consistent with the estimation-noise correction the adjustment makes. Rejection supports incremental predictive content in the augmented forecast, which the leave-group-out attribution locates in the surface variables. It does not guarantee that the augmented model lowers realised forecast error at this sample size, which is why the raw comparison remains inconclusive and the day-level claim is stated as positive rather than established. The placebo and seed analyses summarised in Section 5 are part of the same validation layer. Three independent permutations of the derivatives day-mapping, re-fitted from scratch under the locked M5 configuration and the identical scaling and data protocol, give pooled AUCs of 0.6307 to 0.6327 against the real model's 0.6318, and per-seed within-day MLP information-set gains vary by a factor of two across seeds (-0.0031 , -0.0061 , -0.0033), which is why all main MLP results are three-seed ensembles.

Appendix G Horizon Sweep, VRP and Sequence Model

To test whether the 10-second null reflects the prediction horizon rather than the information set itself, an information-set sweep rebuilds the target at 10 seconds, 1 minute, 5 minutes and 30 minutes and compares the six-feature surface set and the eleven-feature surface-plus-flow set against the order book alone. Within-day increments are significantly negative for both sets at 10 seconds and positive only for the wider set at 1 minute (+0.0010). At 5 minutes the surface set is borderline at +0.0028, falling just short of significance, while the wider set is negative and insignificant, and neither set is significant at 30 minutes (Table 19).

Table 19: Information-Set Sweep by Prediction Horizon. Within-day AUC for the order book baseline and for the six-feature surface set (D6) and eleven-feature surface-plus-flow set (D11), with within-day increments over the baseline and day-bootstrap confidence intervals. Derivatives levels are the baseline plus the increment. The target is rebuilt at each horizon. The increments are plotted in Figure 4.

Horizon	LOB	D6 Increment [95 % CI]	D11 Increment [95 % CI]
10 s	0.6105	-0.0010 [-0.0019, -0.0002]	-0.0017 [-0.0027, -0.0008]
1 min	0.6096	+0.0005 [-0.0001, +0.0011]	+0.0010 [+0.0001, +0.0020]
5 min	0.6176	+0.0028 [+0.0000, +0.0056]	-0.0028 [-0.0119, +0.0052]
30 min	0.6294	+0.0104 [-0.0062, +0.0276]	-0.0009 [-0.0193, +0.0157]

To determine whether the apparent variance-risk-premium effect comes from implied volatility or from the running-volatility component, the attribution analysis separates the two. Running volatility has positive point estimates at all three attribution horizons, +0.0016, +0.0026 and +0.0152 within-day, significant at 10 seconds and 2 hours. The clean incremental test adds ATM implied volatility alongside running volatility rather than in place of it, and its within-day increment is -0.0000 at 10 seconds, -0.0013 at 30 minutes and -0.0061 at 2 hours, none of them significant. The difference feature replaces running volatility rather than adding to it, so the comparison tests functional form rather than an isolated implied-volatility effect. In that difference form the 30-minute increment over the running-volatility model is +0.0089 with an interval of [+0.0026, +0.0157], a conventionally significant positive estimate within this attribution analysis, while supplying implied volatility as a separate column beside running volatility adds nothing detectable at the same horizon. An earlier version of the running-volatility feature backfilled the opening bins of each session and produced an apparently significant 30-minute effect in this comparison. Correcting that leakage removed it and a later preprocessing correction returned it, so the difference-form estimate is unstable across data-handling choices, and at 2 hours the same difference form is significantly worse than plain running volatility at -0.0471 [-0.0893, -0.0074], while the clean incremental test is null throughout (Table 20 Panel A).

A Cboe VIX cross-check tests whether a more recent implied-volatility reading would help at 30 minutes. Replacing the end-of-day ATM-IV input of the 30-minute variance-risk-premium feature with the same-morning VIX open makes the feature slightly worse (-0.0072, not significant), even though the VIX open is measured hours later and is the stronger same-day volatility nowcast (out-of-sample R^2 0.704 against 0.626 for prior-day ATM implied volatility). This argues against the timing of the end-of-day surface as the explanation for the weak intraday implied-volatility channel (Table 20 Panel B).

The temporal convolutional network reads rolling 30-bin (5-minute) windows of the LOB+ features (three convolutional layers, about 7,000 parameters, three-seed ensembles, early stopping on validation AUC, train-only scaling). It underperforms the boosted trees on the same features (within-day 0.5732 against 0.6136) and the derivatives increment in the sequence class is negative (pooled 0.6128 to 0.6077, within-day -0.0134 between seed ensembles). The 5-minute sequence representation adds nothing at the 10-second horizon, consistent with signed order flow summarising much of the relevant recent history (Table 20 Panel C).

Table 20: Variance-Risk-Premium Attribution, IV Timing and Sequence Model. Panel A separates running volatility from implied volatility. Panel B replaces prior-day IV with the same-morning VIX open. Panel C reports three-seed temporal convolutional network ensembles. Panels A and B report within-day AUC increments with day-bootstrap confidence intervals.

Panel A: Within-Day Increments			
Horizon	Running Vol vs LOB	Difference Form	IV Column Added
10 s	+0.0016 [+0.0005, +0.0025]	−0.0001 [−0.0010, +0.0009]	−0.0000 [−0.0002, +0.0001]
30 min	+0.0026 [−0.0003, +0.0055]	+0.0089 [+0.0026, +0.0157]	−0.0013 [−0.0040, +0.0008]
2 h	+0.0152 [+0.0029, +0.0275]	−0.0471 [−0.0893, −0.0074]	−0.0061 [−0.0177, +0.0049]
Panel B: IV Timing, 30 Minutes			
IV Input	Increment [95 % CI]	Same-Day Vol OOS R^2	
Prior-day OM ATM IV	+0.0089 [+0.0026, +0.0157]	0.626	
Same-morning VIX open	+0.0017 [−0.0013, +0.0050]	0.704	
Timing effect	−0.0072 [−0.0145, +0.0001]		
Panel C: Temporal Convolutional Network, Three-Seed Ensembles			
Information Set	Pooled AUC	Within-Day AUC	
LOB+	0.6128	0.5732	
LOB+ and Deriv	0.6077	0.5598	
Increment	−0.0051	−0.0134	

Appendix H Analysis Code

The analysis is implemented in one executed Jupyter notebook spanning Sections 0 to 22 with 54 code cells, supported by two short scripts reproduced inside it, one building the order-flow-imbalance files and one counting the residual wide-spread rows quoted in Appendix B. The notebook reprocesses the raw Databento and OptionMetrics data when the derived per-day caches are absent and otherwise loads them. Together these form the code record and produce all reported results under the pinned environment of Appendix C. The Appendix B statistics and that wide-spread count come from the stored feature grid restricted to the primary test window, and the two preprocessing checks of Appendix C, the training-window dispersion cutoff and the vega-unit identity, run in the notebook itself. The notebook and both scripts are hosted in the accompanying code repository at <https://github.com/EC152/2026-AP-RMFE>. Table 21 maps the notebook structure, and three short excerpts below illustrate the methodological corrections central to the results.

Table 21: Structure of the Analysis Notebook

Section	Content
0 to 1	Environment setup, data verification
2 to 4	LOB feature extraction (per-day, resume-safe), OptionMetrics features, merged 10-second grid
5 to 6	Target construction, descriptive statistics, chronological train-validation-test split and scaling protocol
7 to 9	Model training M1 to M8, predictive evaluation, feature importance
10 to 11	Out-of-sample robustness window, results and hyperparameters
12	Diagnostics: ROC, weekly stability, correlation, decile lift, regime and time-of-day conditioning
13 to 14	Probability calibration, decision rule, economic translation
15 to 16	Validation layer: within-day decomposition, placebo permutation, seed dispersion, activity and label-threshold robustness, final corrected results
17 to 19	Horizon sweep, day-level forecasting and inference, VRP attribution, integrity audit, Cboe VIX cross-check and preprocessing verification
20 to 22	Order-flow imbalance and the LOB+ production set, sequence model, final answer and desk recommendation

H.1 Target Construction

The binary adverse-selection target y_t is built once from the causal ten-second grid. Intervals with no net directional signed flow ($q_t = 0$) are assigned $y_t = 0$ by convention, since there is no exposure direction to be adverse to. The stored grid is indexed by each interval's left edge, whereas t throughout this report denotes the corresponding interval close and decision time. The two conventions differ by a constant offset and leave every feature and label window unchanged.

```
def build_target(df, tau=0.5):
    out = df.copy()
    out['mid_next'] = out.groupby('date')['mid'].shift(-1)
    out['mid_change'] = out['mid_next'] - out['mid']
    out['quote_dir'] = np.sign(out['signed_flow'])
    out['y'] = np.where(
        out['quote_dir'] != 0,
        (out['mid_change'] * out['quote_dir']
         >= tau * out['half_spread']).astype(int),
        0,
    )
    return out.dropna(subset=['mid_next'])
```

H.2 Consistent Scaling for Final Predictive Models

The final predictive models are trained on combined training-plus-validation data, so every matrix they are subsequently evaluated on must be transformed with a scaler fitted on that same window. An early version of the pipeline evaluated the final models against test and validation matrices scaled with a train-only scaler instead, silently distorting every downstream number (Section 6).

The excerpt below is the corrected scaling block. The economic tier instead uses models fitted on the training window only, with train-only scalers, so that validation predictions are out of sample for calibration and threshold selection.

```
scaler_lob_tv = StandardScaler()
X_tv_lob = scaler_lob_tv.fit_transform(trainval[LOB_FEATURES])

# Final models are trained on train+val-scaled features, so every
# matrix they are evaluated on must use the same train+val
# scalers, not scalers fitted on the training window
# alone (LOB-only XGB: 0.529 -> 0.636).

X_va_lob_f = scaler_lob_tv.transform(val[LOB_FEATURES])
X_te_lob_f = scaler_lob_tv.transform(test[LOB_FEATURES])
```

H.3 Placebo Permutation Test

Before accepting an apparent AUC gain from adding derivatives features as genuine, the day-to-derivatives mapping is permuted so the added features carry no real information, and the affected XGBoost model is re-fitted from scratch using the locked M5 configuration under identical scaling and data handling. If permuted noise reproduces the same gain, that gain cannot be attributed to genuine derivatives information. Three permutations are run in the notebook, and the variance across them is small.

```
for perm in range(3):
    shuffled = deriv_by_day.copy()
    shuffled[:] = deriv_by_day.sample(
        frac=1.0, random_state=perm).values
    plc = full_d.copy()

    plc[DERIV_FEATURES] = shuffled.reindex(day_key.values).values

    p_tr = plc[plc['date'].astype(str) <= TRAIN_END]

    p_va = plc[(plc['date'].astype(str) >= VAL_START) &
               (plc['date'].astype(str) <= VAL_END)]
    p_te = plc[plc['date'].astype(str) >= TEST_START]

    p_tv = pd.concat([p_tr, p_va], ignore_index=True)

    sc_tv_p = StandardScaler().fit(p_tv[ALL_FEATURES])

    mp = XGBClassifier(**params5, subsample=0.8, n_estimators=ep5,
                       scale_pos_weight=sp_full,
                       random_state=42, verbosity=0)
    mp.fit(sc_tv_p.transform(p_tv[ALL_FEATURES]), p_tv['y'].values)

    pp = mp.predict_proba(
        sc_tv_p.transform(p_te[ALL_FEATURES]))[:, 1]
```